Software Requirements Specification: UTeM Micro-Credential Management System (MCMS)

Oct 23, 2026 · @adomin

This document states what MCMS must do, for whom and under which rules, so the development team can design, build and test it. It is a Software Requirements Specification (SRS): the standard document that describes **what** a system shall do. It also holds an architecture overview and a data model so developers see the shape of the solution, but detailed technical design (code structure, screens, database scripts) belongs in a later design document.

Item	Detail
Document title	Software Requirements Specification, UTeM Micro-Credential Management System (MCMS)
Version and status	0.1, draft for review
Date	23 October 2026
System owner	Centre for Open and Distance Learning (CODL), Universiti Teknikal Malaysia Melaka (UTeM)
Intended readers	Developers, testers, architects, product owner
Source of rules	Garis Panduan Micro-Credential UTeM (Bab 1–18, Lampiran A–H). The file name says version 2.0 final, 15 September 2026, while its cover page says version 1.0, 18 September 2026. CODL should confirm which one is the controlled version.
Structure followed	ISO/IEC/IEEE 29148 (requirements engineering), adapted for this project

1. Introduction

1.1 Purpose

This SRS defines the functional, data, interface, quality and rule requirements of MCMS. It is the agreed basis for design, development, testing and acceptance. Every requirement carries an ID so it can be traced to the Garis Panduan clause it comes from and to a test.

1.2 Scope

MCMS is a **new, standalone** system that manages the whole life of a UTeM Micro-Credential (MC): from proposal and approval, through offering, learning results and credentials, to Academic Credit Record (ACR), Academic Bank of Credit (ABC) deposit tracking, credit transfer, stackable degree pathways, monitoring and continuous quality improvement (CQI).

In scope	Out of scope
All three categories: UMC, SAMC, MCE	Delivering learning content (done on UCreds)
Proposal, review, approval and versioning	The national ABC platform and MQA/QVC procedures (MCMS prepares and tracks, UTeM staff submit)
Registration, enrolment, results, completion	The full student information system and degree programme rules
Digital Badge, Certificate, Digital Credential and public verification	The payment hub itself (a dummy service stands in for version 1)
QV evidence for SAMC, monitoring, CQI, change control	Partner-delivered micro-credentials (possible later)
ACR, ABC tracking, single redemption, credit transfer, stackable pathway checks	Open public sign-up (UTeM account holders only in version 1)
Audit trail, records, reports	

1.3 Intended readers

Reader	Read first
Developers	Sections 3, 5, 7, 8
Testers	Sections 3, 7, 10
Architects and security	Sections 2, 4, 5, 6
CODL product owner	Sections 1, 2, 7, 11 and the appendices

1.4 Definitions and abbreviations

Term	Meaning
MC	Micro-Credential: certification of a small, focused unit of learning with defined outcomes and assessment
UMC	Unbundled Micro-Credential: made by unbundling an accredited credit course in a UTeM programme; keeps its link to that parent course
SAMC	Stand-alone Micro-Credential: its own offering, not part of an accredited programme; at least 40 hours SLT
MCE	Micro-Credential Eksekutif: non-credit short learning, less than 40 hours SLT
SLT	Student Learning Time: total estimated learning time including activities and assessment
QV / QVC	Quality Verification for SAMC / the committee that carries it out under MQA rules
MQA / MQF	Malaysian Qualifications Agency / Malaysian Qualifications Framework
ACR	Academic Credit Record: the record used in the ABC mechanism

Term	Meaning
ABC	Academic Bank of Credit: central digital depository of ACR
Achievement Record	UTeM's internal record of modules passed, badges and MC completion. It is **not** an ACR.
Digital Badge	Proof of achievement of one module or unit. Not academic credit.
Certificate / Digital Credential	Proof of completing the whole MC. Does not by itself give credit or transfer.
Credit transfer	Receiving programme's evaluation and acceptance of credit. A separate step from accumulation.
Stackable Degree	Planned pathway where eligible credit counts toward part of a degree programme
Single redemption	Credit already used for one qualification cannot be used for another
Material change	Change to outcomes, SLT, assessment, module structure, credit, MQF level, parent course link or QV status
PTj	Pusat Tanggungjawab: faculty, school, institute or centre responsible for an offering
CODL	Centre for Open and Distance Learning, owner of MCMS and UCreds
UCreds	UTeM's online learning platform for micro-credentials, MOOC and ODL; the pilot learning platform
HEPA / Pejabat Pendaftar	Student Affairs and Alumni Department / Registrar's Office
RBAC	Role-based access control
SSO	Single sign-on

1.5 References

1. Garis Panduan Micro-Credential UTeM (the Guide), CODL UTeM, with Lampiran A–H.
2. MQA: Guidelines to Good Practices on Micro-Credentials; Quality Verification of Stand-alone Micro-Credentials; Guidelines on the Academic Bank of Credit; Malaysian Qualifications Framework (second edition) and current amendments.
3. Guidelines on Developing Academic Programme through Stackable Degree Pathway.
4. MQA COPPA: Open and Distance Learning, where delivery enters ODL scope.
5. UTeM academic regulations, quality assurance policy, and records and document management policy.
6. ISO/IEC/IEEE 29148, requirements engineering.
7. Open Badges 3.0 and W3C Verifiable Credentials data model (credential format guidance).
8. OWASP Application Security Verification Standard; WCAG 2.1; Personal Data Protection Act 2010 (Act 709), with current amendments.

1.6 Conventions

- **ID format:** `AUTH-03` is a requirement; `BR-12` is a business rule; `T07` is an acceptance test; `A5` is an assumption.
- **Wording:** *shall* is mandatory. *Should* is strongly desired. *May* is optional.
- **Priority: M** must have, **S** should have, **C** could have.
- **Phase:** delivery phase 1–5 (section 11).
- **Source tag:** `[G 8.7.3]` means Guide clause 8.7.3; `[App E2]` means Guide appendix E, table E2.
- **Configurable value:** a number or list that sits in the settings table, never hard-coded.

2. Overall description

2.1 Product perspective

MCMS is new and stands on its own, with its own database, login handling and workflow. It does not extend UCreds or the student information system (SIS). It talks to them, and to other systems, only through adapters. The diagram shows who uses MCMS and which systems it connects to.

2.2 Architecture overview

MCMS is a **modular monolith**: one deployable application split into clearly separated modules (M1–M16) that call each other through internal interfaces. Everything outside the application is reached through an adapter, so a stub or manual version can be replaced by the real connector later without touching the modules. The system runs on UTeM's own servers.

2.3 The central idea: separate recognition stages

The Guide treats each of the following as a **different thing** and warns against merging them [G 12–14, 18.14]. The system shall keep them as separate records with separate decisions, so none can be skipped or mistaken for another. This is the single most important rule for developers to understand.

2.4 User classes

Code	User class	Description and main tasks
P1	Coordinator, developer, facilitator	Create proposals, design learning, run cohorts, enter results, raise CQI. Limited to own PTj.
P2	PTj / faculty	Verify category and review, own the offering, confirm readiness, approve credentials, decide individual credit transfer cases
P3	Academic authority (Senate)	Approve offerings, credit transfer rules, stackable mapping, material changes
P4	Quality assurance	Review, monitor, QV evidence, CQI
P5	Platform and technical support	Enrolment set-up, UCreds sync, integration monitoring
P6	Records and credential unit (assumed CODL)	Issue and correct credentials, prepare ACR, maintain records
P7	ABC depositor	Deposit eligible ACR to ABC and record status

Code	User class	Description and main tasks
P8	Participant	Sign in, enrol, learn, view results and credentials, give consent, apply for transfer
X1	Reviewer or assessor	Review a proposal objectively; declares conflict of interest
X2	Industry expert / SME	Gives input on competency and relevance only
X3	Verifier	Employer or public checking a credential; no login
X4	Receiving programme	Evaluates transferred credit
S1	System administrator (CODL)	Users, roles, templates, settings; never approves academic matters

2.5 Operating environment

- UTeM on-premises servers, containerised or virtual machines, separate development, test and production.
- Participant and staff access through current desktop and mobile browsers.
- Pilot learning platform: UCreds. Staff login through UTeM single sign-on.
- Offices and data owners: CODL, PTj, Pejabat Pendaftar, HEPA, UTeM ICT.

2.6 Design and implementation constraints

ID	Constraint
C1	The system shall be built as a new standalone system, not an extension of UCreds or SIS.
C2	All data shall stay on UTeM-controlled infrastructure.
C3	Every external system shall be reached through an adapter interface with a stub or manual implementation for version 1.

ID	Constraint
C4	Thresholds (40 hours SLT, 3-year QV limit, 10-year credit validity, 30% home credit, residency, alert days, retention) shall be settings, not code.
C5	Credit, MQF level and QV wording shall be generated only from verified records, never typed by hand.
C6	Approvals, credential actions, results changes and corrections shall be written to an append-only audit log.
C7	The interface and credentials shall support Bahasa Melayu and English.
C8	Credentials shall use an open, signed format; blockchain is not required.
C9	A UMC module is never a credit unit; only the whole MC carries credit.

2.7 Assumptions and dependencies

If an assumption proves wrong, the effect noted in the right-hand column applies. Items marked *confirm* are open questions for CODL.

ID	Assumption	Effect if wrong
A1	The Senate approves offerings, transfer rules and stackable mapping. Faculty decide each credit transfer case.	Approver bodies are settings; workflow routing changes, no rebuild. *Confirm.*
A2	ABC and QVC/MQA accept submissions manually in version 1.	A real adapter is added later.
A3	UTeM's payment hub is not ready; a dummy payment service is used.	Replace one adapter.
A4	UCreds is the pilot learning platform. Its connection method (web service, data export, Moodle link) is not yet known.	CSV import is the fallback; connector design shifts. *Confirm.*

ID	Assumption	Effect if wrong
A5	UCreds already issues module badges. MCMS imports and checks them and issues certificates and Digital Credentials.	If UCreds should not issue badges, MCMS issues them instead. *Confirm.*
A6	Outside applicants register through the Pejabat Pendaftar or HEPA, receive a UTeM account, and are recognised by an account attribute or imported list.	Add an applicant pre-registration screen. *Confirm.*
A7	The records and credential unit (P6) and the product owner are CODL.	Role holders change in settings.
A8	Retention periods in section 6 are placeholders.	Edit settings.
A9	Sizing: up to 200 offerings and 20,000 participants a year, peaks of 2,000 registering within hours.	Capacity plan changes.
A10	Launch date, budget and team size are not known.	Delivery estimates in section 11 are rough only.

2.8 Product functions

Module	Function	Main Guide source
M1	Identity, roles and access	Bab 15–16, 18.11, App H
M2	Category determination	6.5
M3	Proposal and design workspace	Bab 9, App A
M4	Review and approval workflow	9.4–9.5, 15.4
M5	Offering and catalogue	9.6, 10.3.1, App B
M6	Registration, enrolment, cohorts	10.3
M7	Learning platform connection	10.1–10.2
M8	Assessment and verification	8.6, 10.4

Module	Function	Main Guide source
M9	Achievement record (internal)	8.7.5, 18.3
M10	Credential issuance and verification	8.7, 10.5, App E
M11	QV management (SAMC)	11.2, App D
M12	ACR, ABC tracking, redemption	Bab 12, App F
M13	Credit transfer and stackable pathway	Bab 13–14, App F–G
M14	Monitoring, CQI, change control	Bab 11, 17, App C
M15	Records, audit, reporting	Bab 18, App H
M16	Notifications and integration hub	Bab 10, 18